REVIEWER SYMBOLS: NAVY - 11 USAF - MT ARMY - MT USER SYMBOLS:

"Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS."

This military standard is approved for use by all Departments & Agencies of the Department of Defense. Selection for all engineering and design applications and for repetitive use should be made from this document.

FED. SUP CLASS 4710

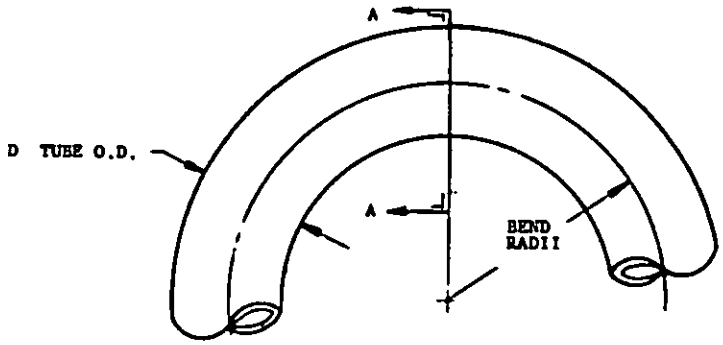
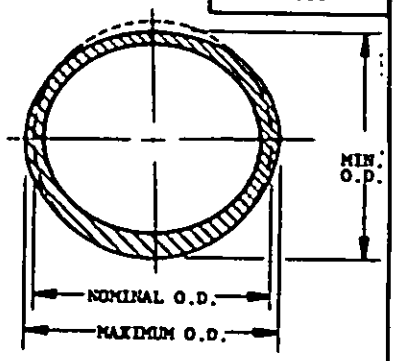


FIGURE 1. TUBE



SECTION A - A

TABLE I - BEND RADII FOR CRES AND ALUMINUM TUBING

TUBE OD	SPECIAL BEND RADII SEE NOTE (5)		RECOMMENDED BEND RADII		ADDITIONAL BEND RADII
	1 - 1/2 D	2D	3D	4D	6D
1/8	.188	.250	.375	.500	.750
3/16	.281	.375	.563	.750	1.125
1/4	.375	.500	.750	1.000	1.500
5/16	.469	.625	.938	1.250	1.875
3/8	.563	.750	1.125	1.500	2.250
7/16	.656	.875	1.312	1.750	2.625
1/2	.750	1.000	1.500	2.000	3.000
5/8	.938	1.250	1.875	2.500	3.750
3/4	1.125	1.500	2.250	3.000	4.500
7/8	1.3125	1.750	2.625	3.500	5.250
1	1.500	2.000	3.000	4.000	6.000
1 - 1/8	1.688	2.250	3.375	4.500	6.750
1 - 1/4	1.875	2.500	3.750	5.000	7.500
1 - 3/8	2.063	2.750	4.125	5.500	8.250
1 - 1/2	2.250	3.000	4.500	6.000	9.000
1 - 5/8	2.438	3.250	4.875	6.500	9.750
1 - 3/4	2.625	3.500	5.250	7.000	10.500
1 - 7/8	2.813	3.750	5.625	7.500	11.250
2	3.000	4.000	6.000	8.000	12.000
2 - 1/4	3.375	4.500	6.750	9.000	13.500
2 - 1/2	3.750	5.000	7.500	10.000	15.000
3	4.500	6.000	9.000	12.000	18.000

FLATNESS LIMITATION:

(1) FLATNESS IN THE AREA OF A TUBE BEND SHALL BE DEFINED BY THE FORMULA.

$$\text{FLATNESS \%} = \frac{\text{MAX. O.D.} - \text{MIN. O.D.}}{\text{NOMINAL O.D.}} \times 100$$

(2) THE BENDS WITH WALL THICKNESS PER TABLE II, WHEN USED IN FLUID SYSTEMS WITH WORKING PRESSURES OF 1500 PSI OR GREATER, SHALL NOT EXCEED FLATNESS VALUES OF:
   a. 5% FOR CRES OR ALUMINUM
   b. 3% FOR TITANIUM, COMMERCIALLY PURE AND COLD WORKED, STRESS RELIEVED.

(3) TUBE FLATNESS FOR FLUID SYSTEMS WITH WORKING PRESSURES LESS THAN 1000 PSI SHALL NOT EXCEED 10% FOR CRES AND ALUMINUM AND 5% FOR TITANIUM.

(4) FLATNESS LIMITATIONS APPLY TO BEND RADII LISTED ON THIS STANDARD. APPLICABILITY TO OTHER BEND RADII SHOULD BE VERIFIED BY TESTING.

(5) USE OF SMALL BENDS (LESS THAN 3D) IN HYDRAULIC, PNEUMATIC AND AIRCRAFT FUEL SYSTEMS REQUIRE THE APPROVAL OF THE PROCURING SERVICE. FOR OTHER FLUID SYSTEMS (LESS THAN 1500 PSI) THE SPECIAL BEND RADII COLUMN MAY BE USED.

THIS IS A DESIGN STANDARD. NOT TO BE USED AS A PART NUMBER.

(A) REVISED & REDRAWN.

APPROVED 5 FEB 58 REVISED (A) 1 APRIL 83

P.A. NAVY - AS Other Cust USAF - 99 ARMY - AV	TITLE TUBE BEND RADII	MILITARY STANDARD MS33611
PROCUREMENT SPECIFICATION	SUPERSEDES	SHEET 1 OF 2

DD FORM 1673-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. PROJECT NO. 4730-0462 PLATE NO. 22080

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:05Z
Check the source to verify that this is the current version before use.

USER SYMBOLS:
REVIEWER SYMBOLS:
NAVY -
USAF - 11
ARMY - MI
"Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS."
This military standard is approved for use by all Departments & Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

TABLE II HIGH STRENGTH 3000 PSI TUBING WALLS AND BEND RADII

NOMINAL TUBE O.D.	TUBE WALL THICKNESS		STANDARD BEND RADII	
	21Cr-6Ni-9Mn	Ti-3Al-2.5V	PREFERRED (4D)	MINIMUM (3D)
1/4	.016	.016	1.00	.75
3/8	.020	.019	1.50	1.12
1/2	.026	.026	2.00	1.50
5/8	.033	.032	2.50	1.87
3/4	.039	.039	3.00	2.25
1	.052	.051	4.00	3.00
1-1/4	.058	.065	5.00	3.75
1-1/2	.070			

NOTE: TUBING WALL THICKNESS LESS THAN .020 INCH REQUIRE THE APPROVAL OF THE PROCURING SERVICE.

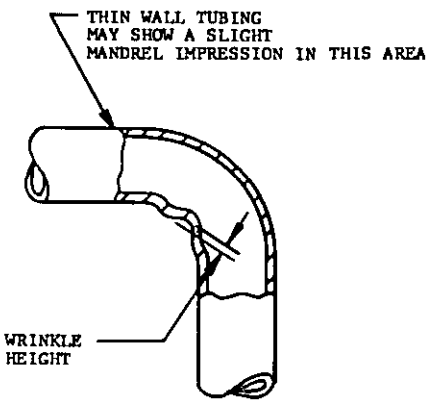


FIGURE 2 INSPECTION AREA

TABLE III PERMISSIBLE WRINKLES AND SCRATCHES IN BENDS

SYSTEM PRESSURE. (PSI)	TUBE MATERIAL	WRINKLE MAXIMUM HEIGHT (% OF TUBE O.D.	SCRATCH MAXIMUM DEPTH (% OF TUBE WALL)
UNDER 500	ALL	2%	10%
500 AND OVER	ALUMINUM ANB CRES	1%	5%
	Ti-3Al-2.5V	NO VISIBLE WRINKLE	5%

APPROVED 5 FEB 58 REVISED (A) FOR CHANGES SEE SHEETS 1 THRU 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:05Z
Check the source to verify that this is the current version before use.